

However, there is one more, third, and probably the most important, reason why functions should not be identified with their formulas.

Let us look at the following expression:

$$y = \begin{cases} \cos x, & x < 0 \\ 1+x^2, & 0 \leq x < 2 \\ \log(x-1), & x \geq 2 \end{cases}$$

How many functions have I defined here?

**Reader:** Three functions: a cosine, a quadratic function, and a logarithmic function.

**Author:** You are wrong. The three formulas ( $y = \cos x$ ,  $y = 1+x^2$ , and  $y = \log(x-1)$ ) define in this case a single function. It is defined on the real line, with the law of numerical correspondence given as  $y = \cos x$  over the interval  $]-\infty, 0[$ , as  $y = 1+x^2$  over the interval  $[0, 2]$ , and as  $y = \log(x-1)$  over the interval  $[2, \infty[$ .

**Reader:** I have made a mistake because I did not think enough about the question.

**Author:** No, you have made the mistake because subconsciously you identified a function with its analytical expression, i.e. its formula. Later on, operating with functions, we shall use formulas rather extensively. However, you should not forget that a formula is not all a function is. It is only one way of defining it.

The example above illustrates, by the way, that one should not identify such notions as the domain of a function and the range of  $x$  on which an analytical expression is defined (i.e. the domain of an analytical expression). For example, the expression  $(1+x^2)$  is defined on the real line. However, in the example above the expression was used to define the function only over the interval  $[0, 2]$ .

It should be emphasized that the question about the domain of a function is of principal significance. It goes without saying that the domain of a function cannot be wider than the domain of an analytical expression used to define the function. But it can be narrower.

**Reader:** Does it mean that a cosine defined, for example, over the interval  $[0, \pi]$  and a cosine <sup>defined</sup> over the interval  $[\pi, 3\pi]$  are two different functions?